

Guillaume Sabiron, PhD

R&D Project Leader

Complex Systems, Mobility & Digital Products

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French citizen | Fluent English | Open to international collaboration and regular travel

R&D Project Leader and Systems Engineer with 15 years of experience in applied research, complex systems and digital-product development. Background spanning mobility, energy, aerospace, control engineering and applied AI. Leads multidisciplinary activities involving more than 20 contributors, from technical-roadmap definition through integration, demonstration and operation of robust, operational solutions.

WORK EXPERIENCE

- Jan. **R&D Project Leader**, [IFPEN](#), Solaize, France
2021–Present **Connected mobility, air quality, applied AI and digital products**
- Lead activities involving more than 20 engineers, researchers, PhD candidates and developers across modelling, machine learning, software and data engineering.
 - Define technical roadmaps, priorities, resources, milestones, interfaces and risk-mitigation actions across concurrent projects.
 - Coordinate the development of digital platforms combining physical models, mobility data, artificial intelligence, databases, APIs and visualisation.
 - Oversee **PLANET'AIR**, a platform for analysing and simulating emissions, air quality and noise, and **Geco Air**, a mobility application used by more than 70,000 people over more than 80 million kilometres.
 - Lead technical workshops, project reviews, demonstrations and engagements with industrial, institutional, academic and European partners.
- Oct. **Control Systems Research Engineer**, [IFPEN](#), Solaize, France
2014–Jan. 2021
- Conducted applied research in connected mobility, wind energy and wave-energy conversion, combining modelling, control, sensing, signal processing and software.
 - Designed and experimentally validated Model Predictive Control strategies, including wave-basin testing at Aalborg University during the WEC Control Competition won by the IFPEN team.
 - Worked on wind LiDAR sensing, wind-field simulation and data-driven methods for connected vehicles and pollutant-emission assessment.
 - Used MATLAB/Simulink and Python for modelling, simulation, control design and performance analysis; supervised engineers, interns and PhD candidates.
- Sept. **Research Engineer and PhD Candidate in Bio-inspired Robotics**, [ONERA](#) / [ESA](#) / [AIRBUS](#)
2011–Sept. 2014 / [CNRS](#), France
- 2014 **Vision-based Guidance, Navigation and Control for autonomous lunar landing**
- Designed a GNC architecture combining optic-flow sensing, navigation, spacecraft dynamics and closed-loop control.
 - Developed and verified algorithms through MATLAB/Simulink software-in-the-loop simulations and system-performance analyses.
 - Developed and integrated optic-flow sensor prototypes tested in laboratory conditions and onboard an unmanned helicopter.
 - Spent six months at ESA-ESTEC within the former TEC-ECN section and contributed to technical reviews, publications and international conferences.
- 2012–2013 **Teaching Activities, MSc level**, *ISAE-SUPAERO and Aix-Marseille University*, France
Dynamic-system modelling, control, system identification, signal processing and filtering.
- 2011 **Control Systems Engineer Trainee**, *LSIS, Aix-Marseille University*, Marseille, France
Designed and implemented a robust flight-control law for an unmanned aerial vehicle.
- 2008 **Research Trainee**, *MICA-CNRS*, Hanoi, Vietnam
Developed an open-source GIS/GPS localisation solution.

EDUCATION

- 2011–2014 **PhD in Bio-inspired Robotics**, *ISAE-SUPAERO*
Vision-based Guidance, Navigation and Control for autonomous aerospace systems, conducted with ONERA, ESA, Airbus Defence and Space and CNRS.
- 2010–2011 **Specialisation Semester in Control Theory**, *LTH*, Lund, Sweden
- 2006–2011 **Engineering Degree, MSc**, *Grenoble INP – ESISAR*, Valence, France
Electronics, computer science and advanced control. Ranked first in my year for three years, graduating with honours.

AREAS OF EXPERTISE

- R&D leadership Roadmaps, prioritisation, work-package definition, resource planning, risk management, multidisciplinary coordination and delivery oversight.
- Complex systems System architectures, multidisciplinary interfaces, dynamic modelling, engineering constraints, technical trade-offs, integration, verification and validation.
- Control systems Guidance, Navigation and Control, robust control, Model Predictive Control, system identification, signal processing, filtering and experimentation.
- AI, data & software Applied machine learning, Python, data pipelines, APIs, databases, visualisation and operational delivery of scientific digital products.
- Partner engagement Needs analysis, stakeholder alignment, technical workshops, demonstrations, project reviews, structured recommendations and international collaboration.

SELECTED PROJECTS

- PLANET’AIR Digital platform for diagnosing, analysing and projecting emissions, air quality and noise for cities, transport infrastructure, airports and ports.
- Geco Air Application and digital services for assessing the environmental impact of travel, combining smartphone data, emission models, APIs and visualisation.
- Wave energy Design and experimental validation of Model Predictive Control strategies to maximise energy capture in wave-energy converters.
- Lunar landing Autonomous architecture combining a bio-inspired optic-flow sensor, navigation, spacecraft dynamics and control for a lunar-landing scenario.
- Wind systems LiDAR sensing, wind-field simulation and modelling and control tools for onshore and offshore wind-energy applications.

SKILLS

- Technical leadership Functional leadership of more than 20 contributors, expert-team facilitation, supervision of engineers, interns and PhD candidates
- Modelling & simulation MATLAB, Simulink, physical-system modelling, algorithm development, software-in-the-loop simulation and experimental validation
- Software & data Python, Pandas, NumPy, Scikit-learn, REST APIs, MongoDB, PostgreSQL, Git/GitLab, CI/CD, Streamlit and Plotly
- Digital products From scientific prototype to operational service, functional architecture, integration, demonstration, deployment and continuous improvement
- Languages French: native English: fluent Spanish: basic
- International Experience in France, Sweden and Vietnam; ESA research placement; European projects, conferences and international expert groups

INTERESTS

- Technology Emerging engineering, simulation, autonomous systems and applied artificial intelligence.
- Activities Paragliding, scuba diving, climbing, snowboarding, hiking and travel.